



APPENDIX D

Bespoke Overcontact Binary Models



All models are wrong, but some are useful.

— *George E. P. Box* —

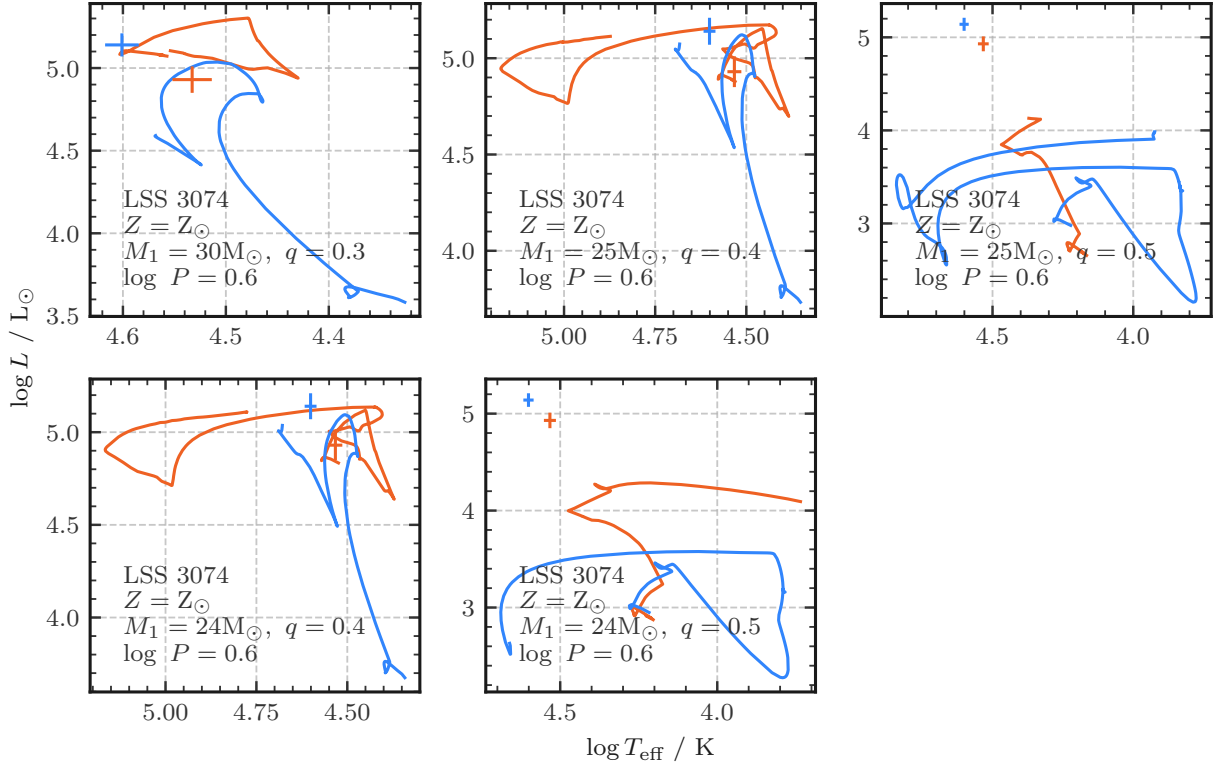


In Table D.1, we list the non-RLOF filling matches to observed contact binaries, contrasted with those that do undergo RLOF as enumerated in Table 5.3. In Figures D.1–D.3, we present the results of our bespoke overcontact binaries as established in Section 5.2. Shown are three parameter configurations, corresponding to the fiducial (normal period) grid in Figure D.1, and our ± 0.1 dex period grid in Figure D.2 (wider periods) and Figure D.3 (shorter periods).

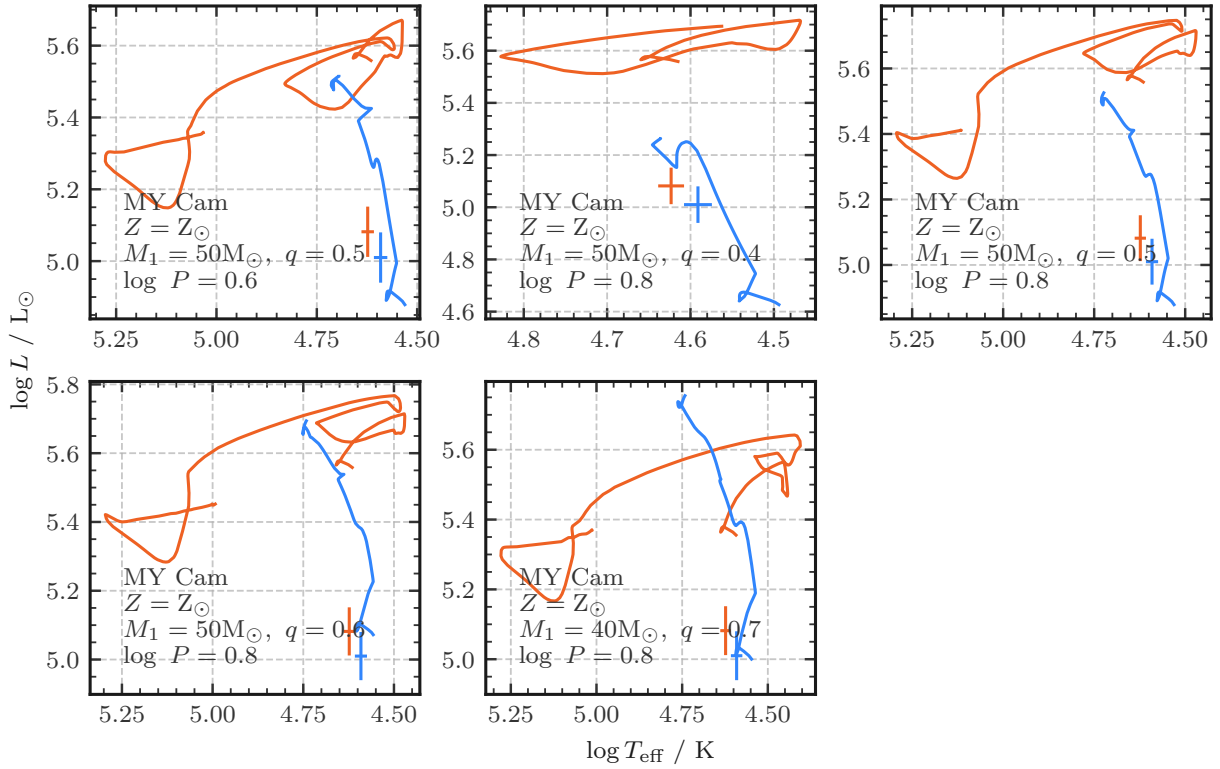
We present the results of our bespoke overcontact binaries as established in Section 5.2. Shown are three parameter configurations, corresponding to the fiducial (normal period) grid in Figure D.1, and our ± 0.1 dex period grid Figure D.2 (wider periods) and Figure D.3 (shorter periods).

System Name	ZAMS parameters				Match Parameters					
	M_1 [$M_\odot$]	M_2 [$M_\odot$]	$\log P$ [d]	M_1 [$M_\odot$]	M_2 [$M_\odot$]	$\log R_1$ [$R_\odot$]	$\log R_2$ [$R_\odot$]	$\log T_1$ [K]	$\log T_1$ [K]	Star 1 RLOF Star 2 RLOF
SMC 108086	20.00	15.00	0.50	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	0.60	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	0.70	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	0.80	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	0.90	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	1.00	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	1.10	20.00	15.00	0.77	0.70	4.53	4.48	✗
SMC 108086	20.00	15.00	1.20	20.00	15.00	0.77	0.70	4.53	4.48	✗
V382 Cyg	35.00	30.00	0.50	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	0.60	29.08	24.58	0.99	0.92	4.57	4.55	✗
V382 Cyg	35.00	30.00	0.60	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	0.70	29.11	24.60	0.99	0.92	4.57	4.55	✗
V382 Cyg	35.00	30.00	0.70	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	0.80	29.04	24.59	1.01	0.93	4.56	4.55	✗
V382 Cyg	35.00	30.00	0.80	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	0.90	29.07	24.60	1.01	0.93	4.56	4.55	✗
V382 Cyg	35.00	30.00	0.90	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	1.00	29.08	24.60	1.01	0.93	4.56	4.55	✗
V382 Cyg	35.00	30.00	1.00	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	1.10	29.09	24.60	1.01	0.93	4.56	4.55	✗
V382 Cyg	35.00	30.00	1.10	35.00	30.00	0.98	0.95	4.58	4.56	✗
V382 Cyg	30.00	25.00	1.20	29.09	24.60	1.01	0.93	4.56	4.55	✗
V382 Cyg	35.00	30.00	1.20	35.00	30.00	0.98	0.95	4.58	4.56	✗

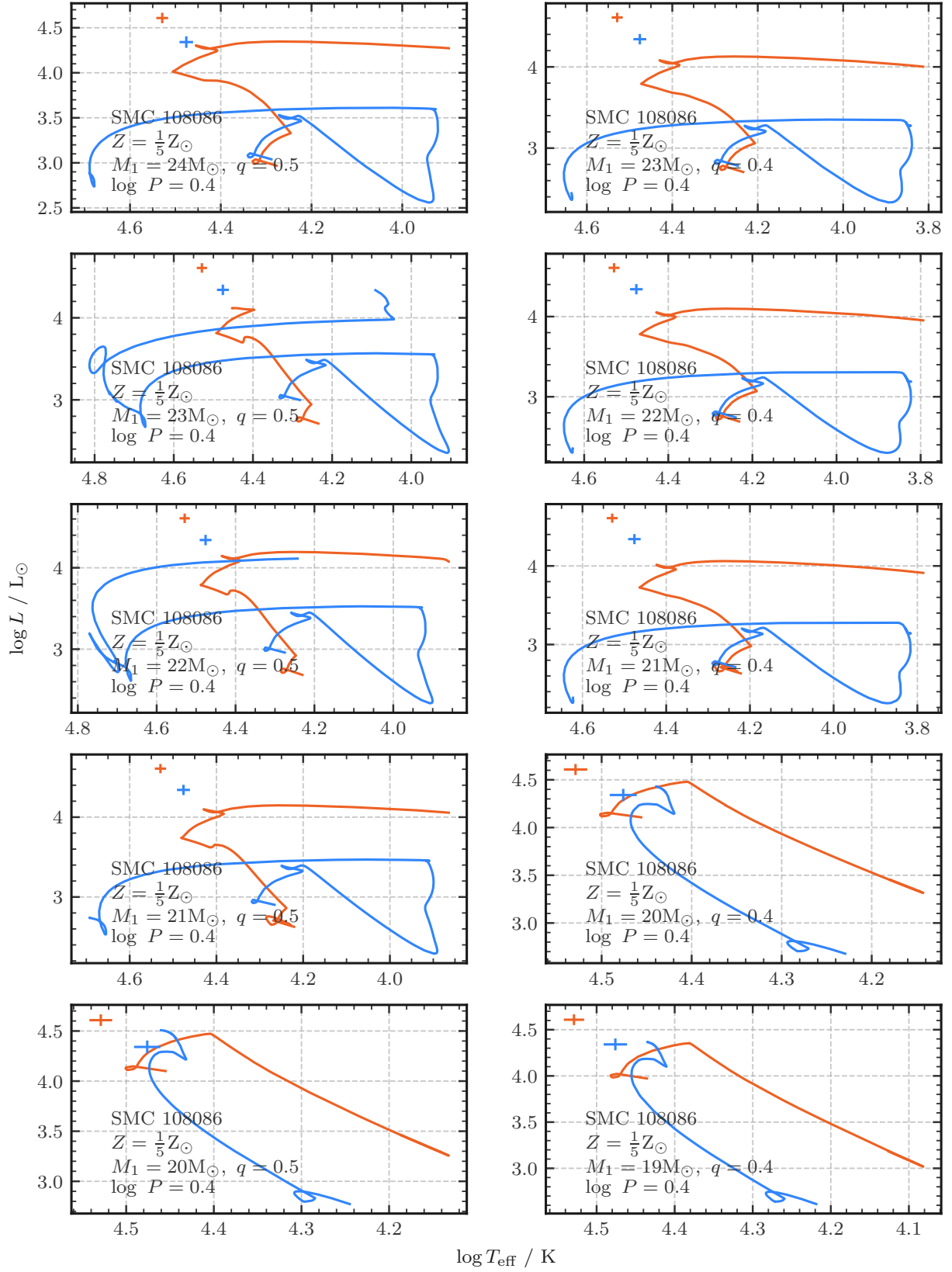
Table D.1: Matches from the TUI code. All parameters are rounded to two decimal places, and are given as at the first moment of matching. Only systems which do not fill their Roche lobes are shown, with the ones filling at least one component shown in Table 5.3.



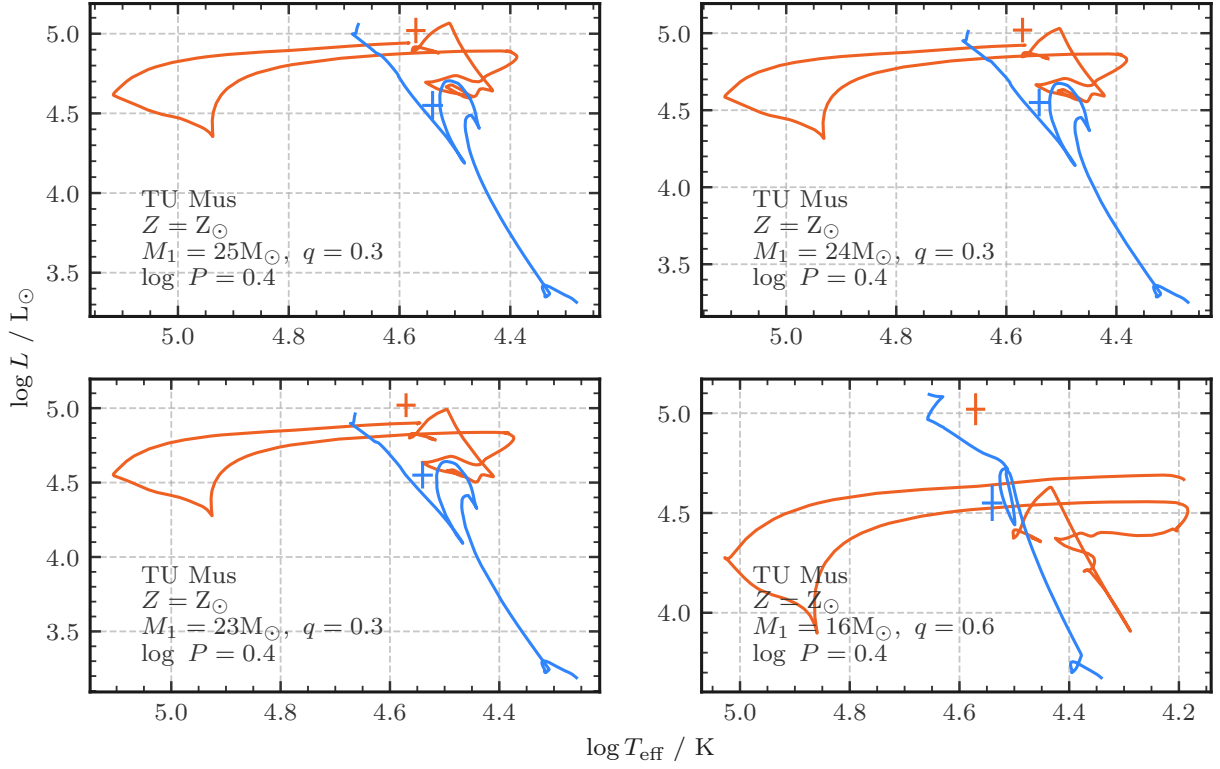
(a) LSS 3074



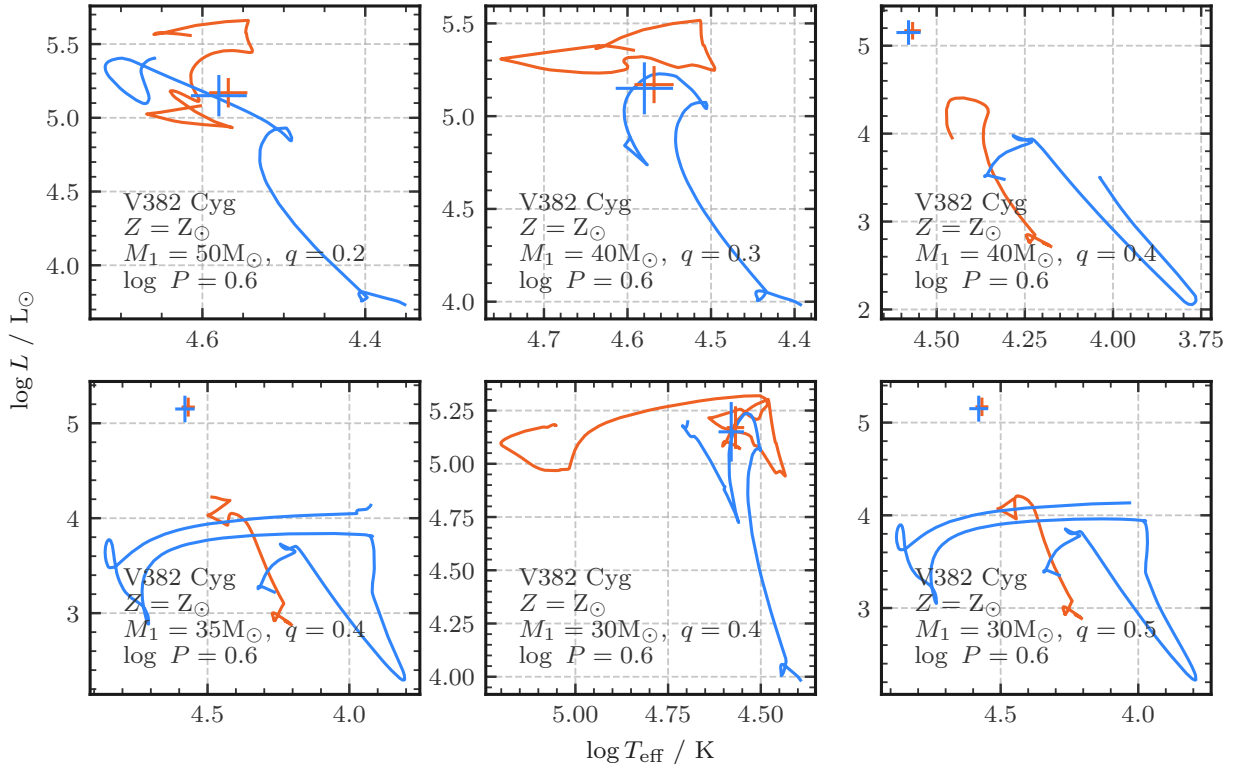
(b) MY Camelopardalis



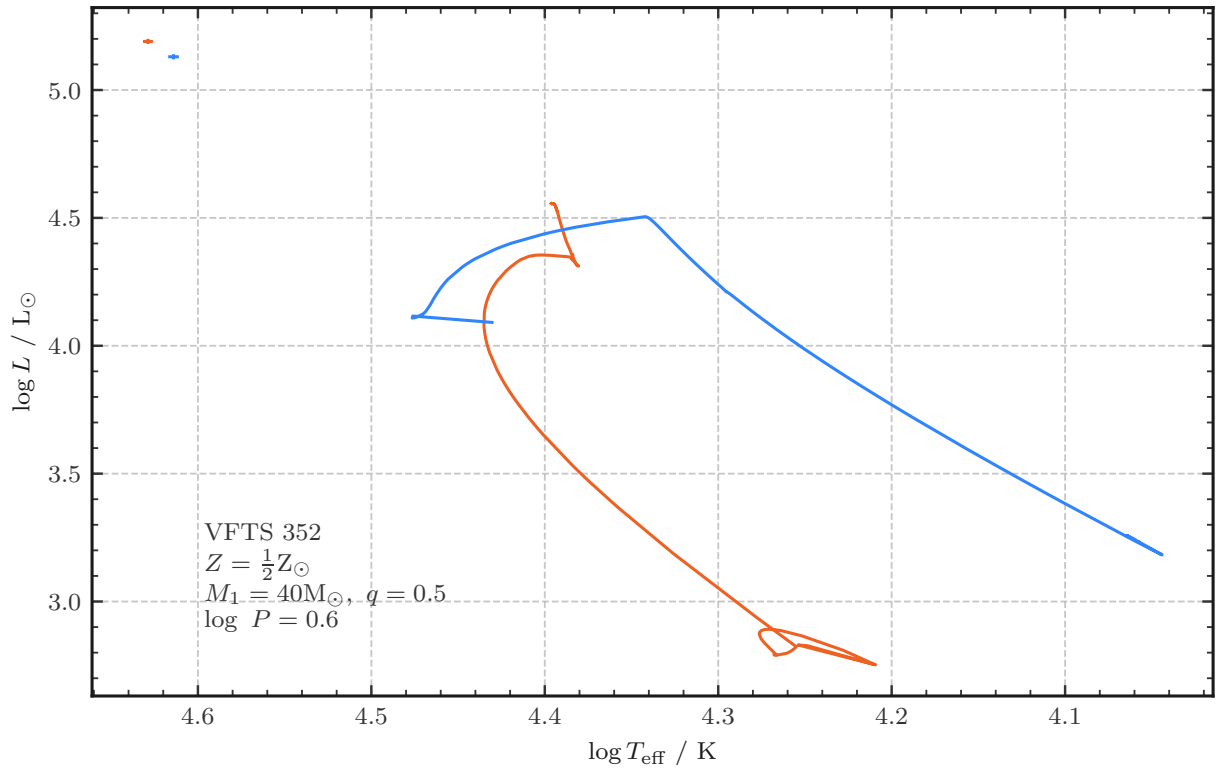
(c) SMC 108086



(d) TU Muscae

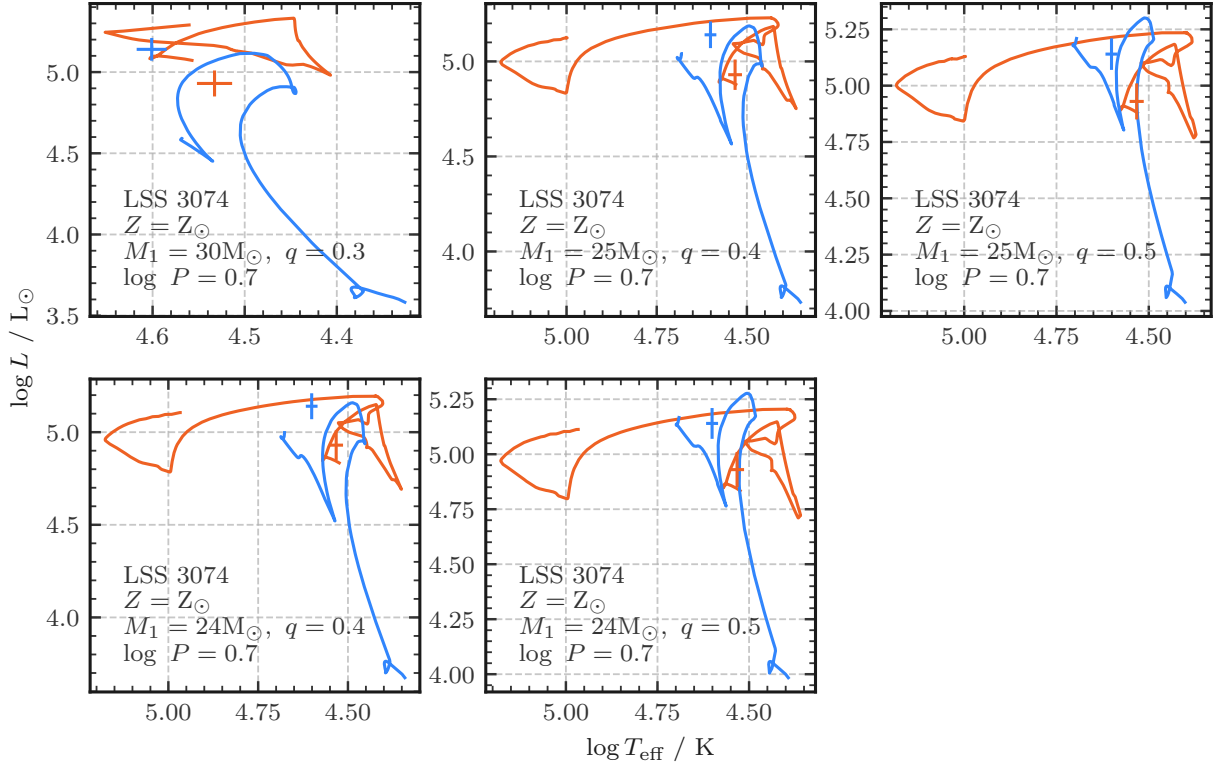


(e) V382 Cygni

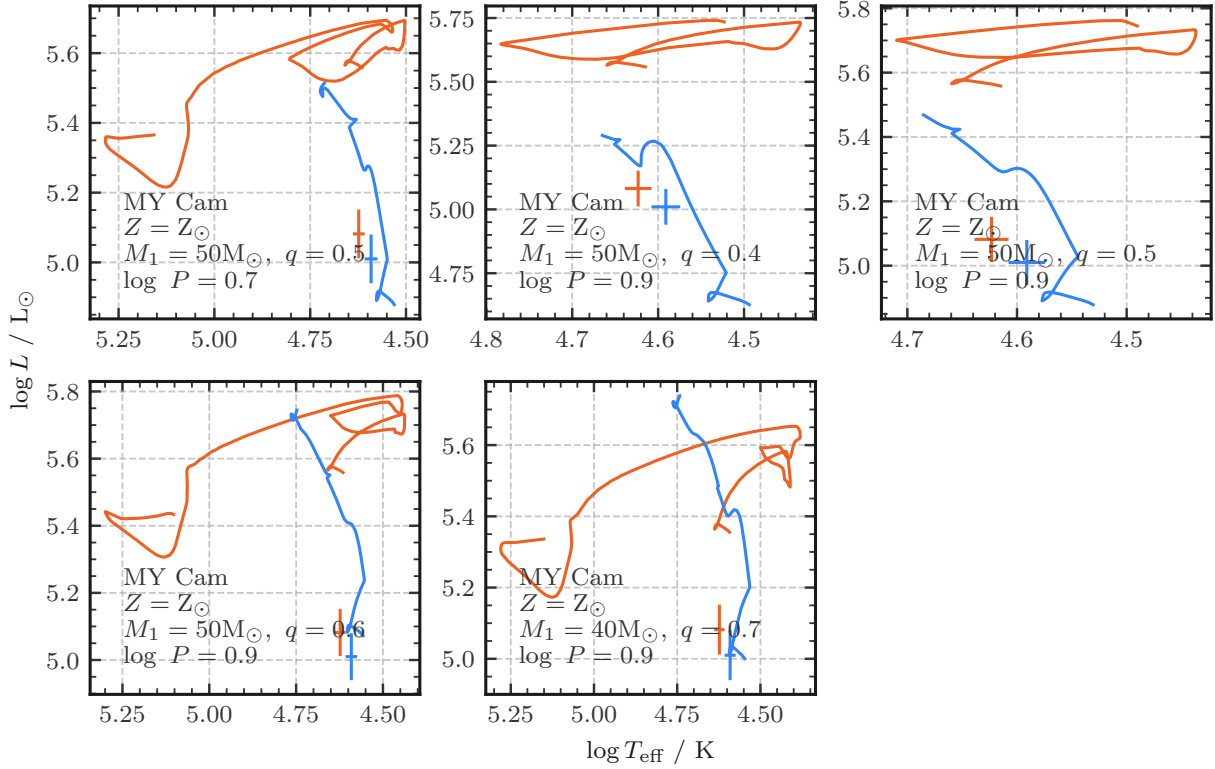


(f) VFTS 352

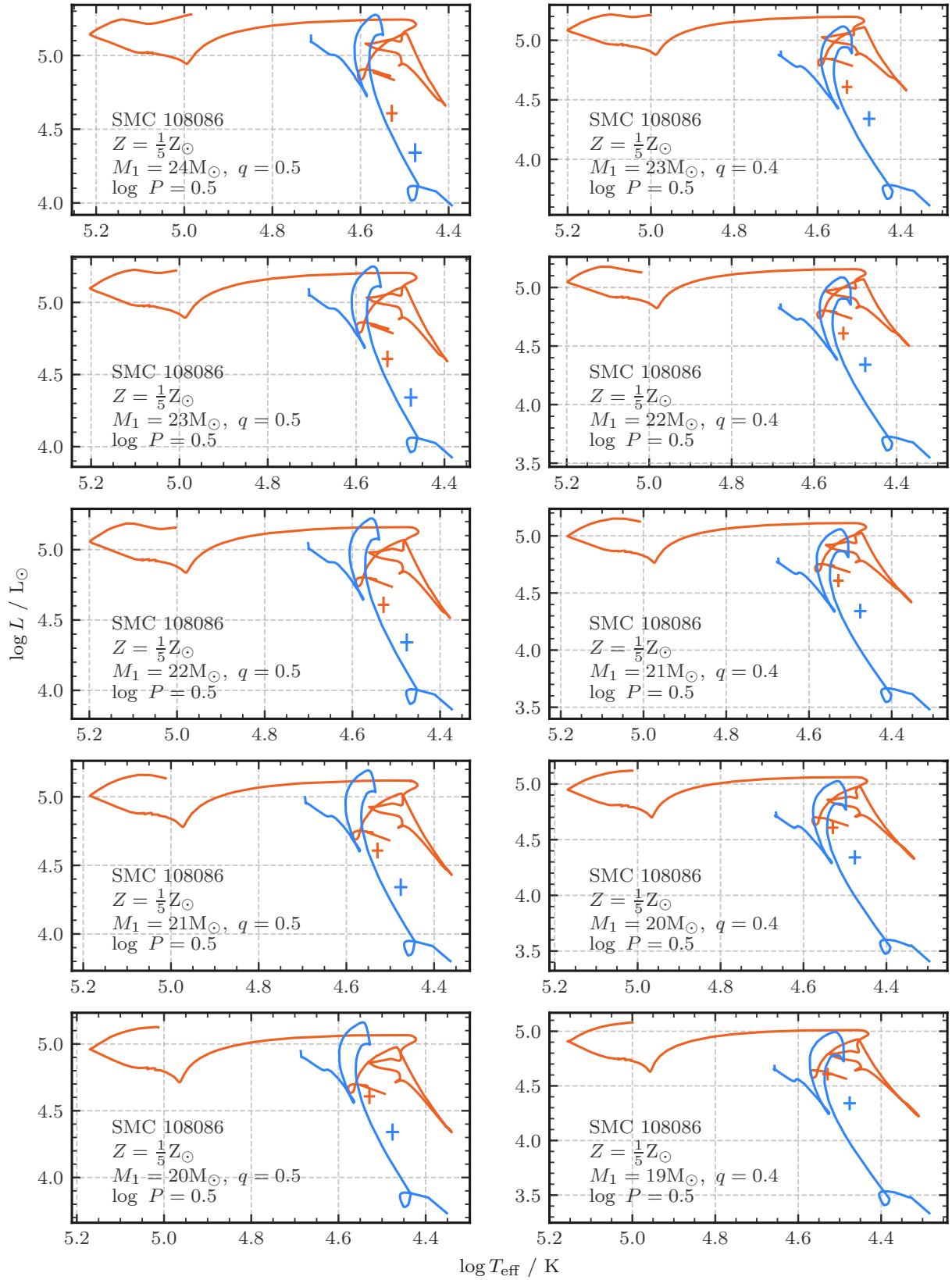
Figure D.1: The results of our bespoke binary evolution for **normal** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.



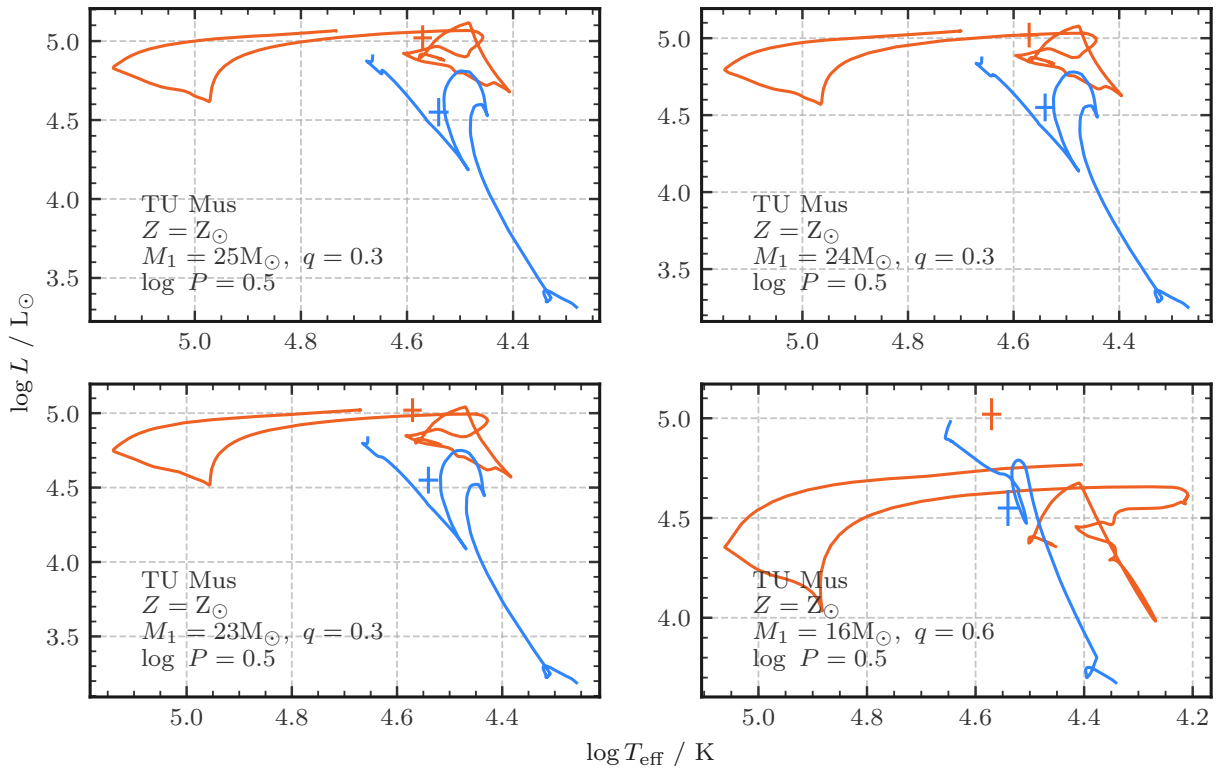
(a) LSS 3074



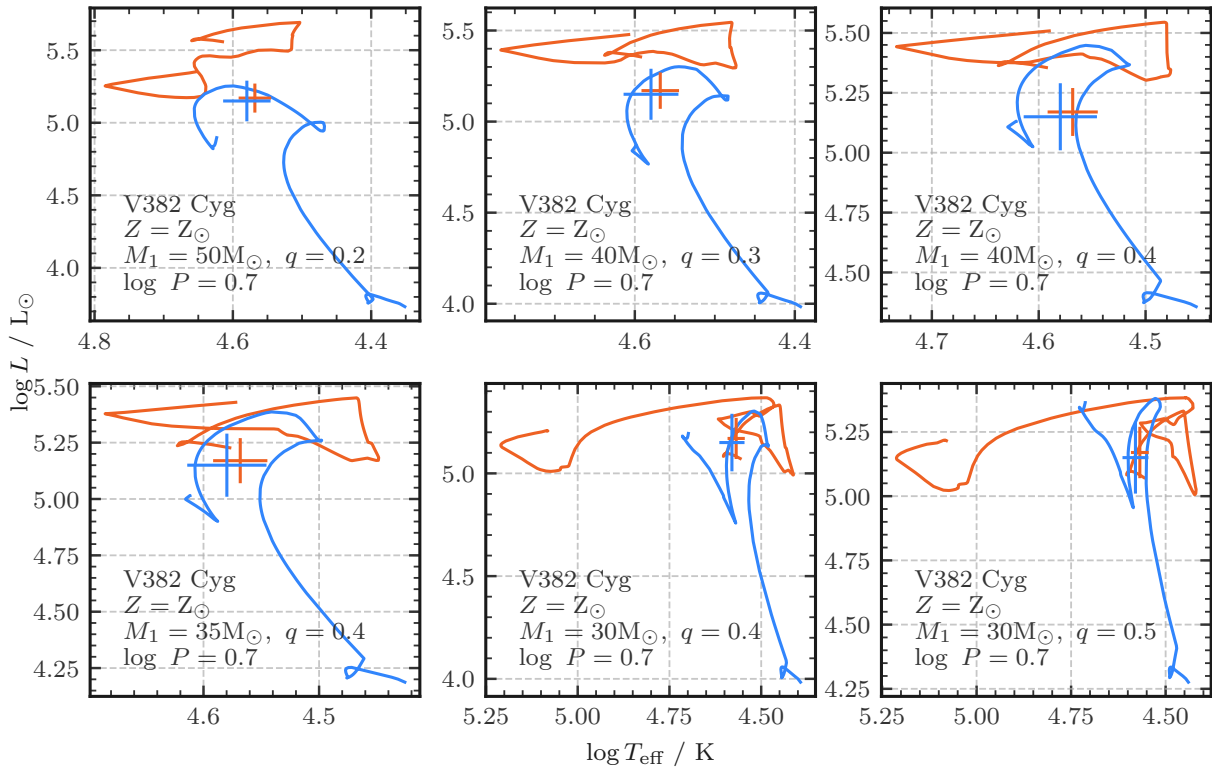
(b) MY Camelopardalis



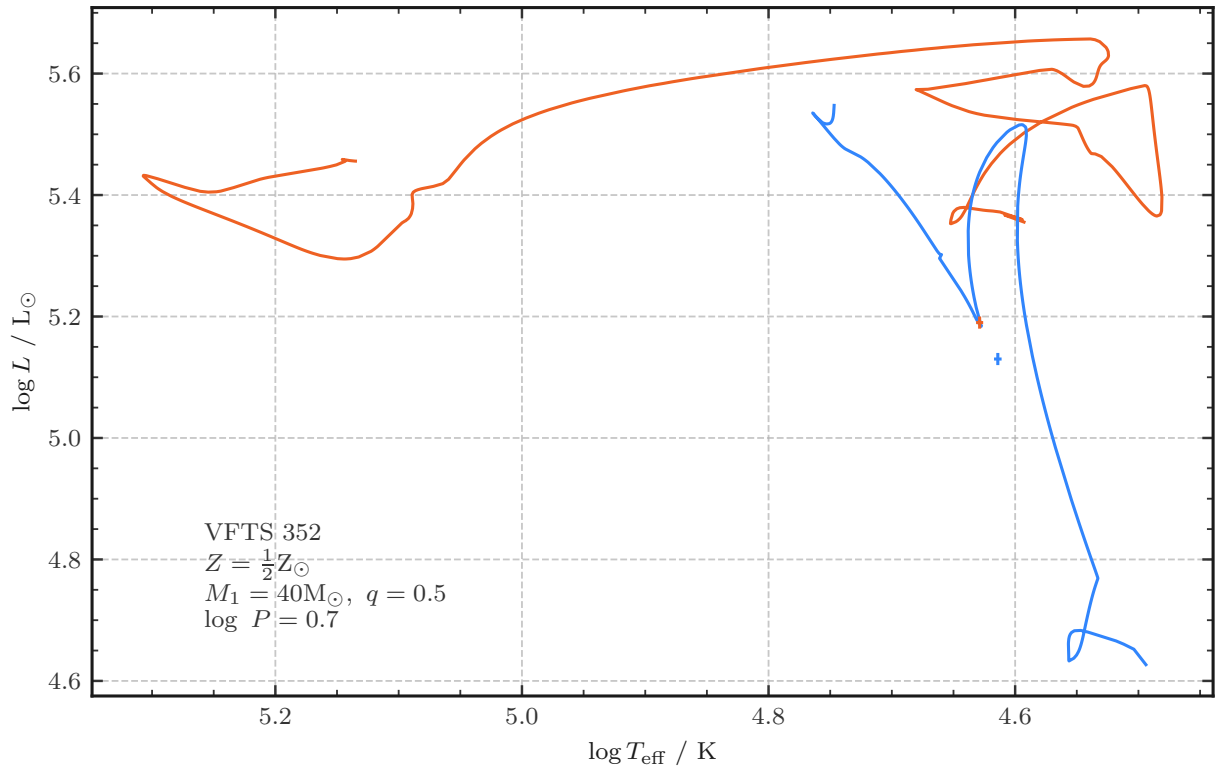
(c) SMC 108086



(d) TU Muscae

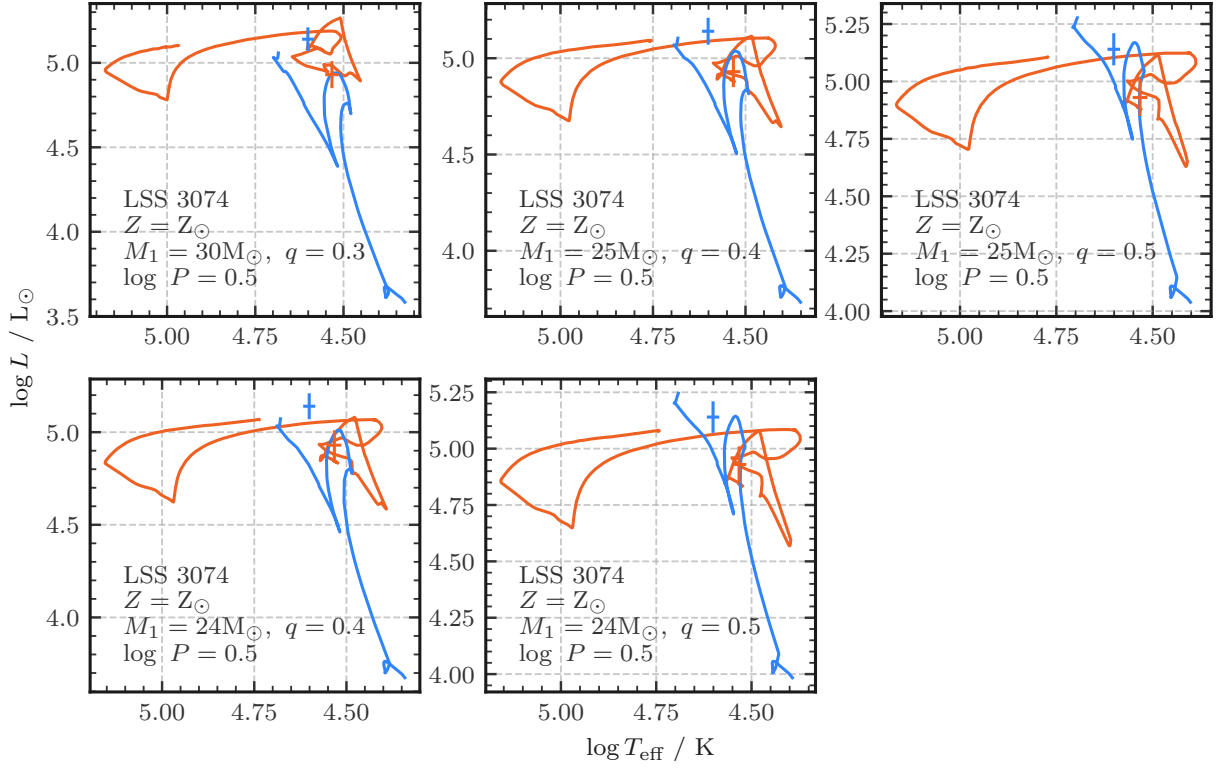


(e) V382 Cygni

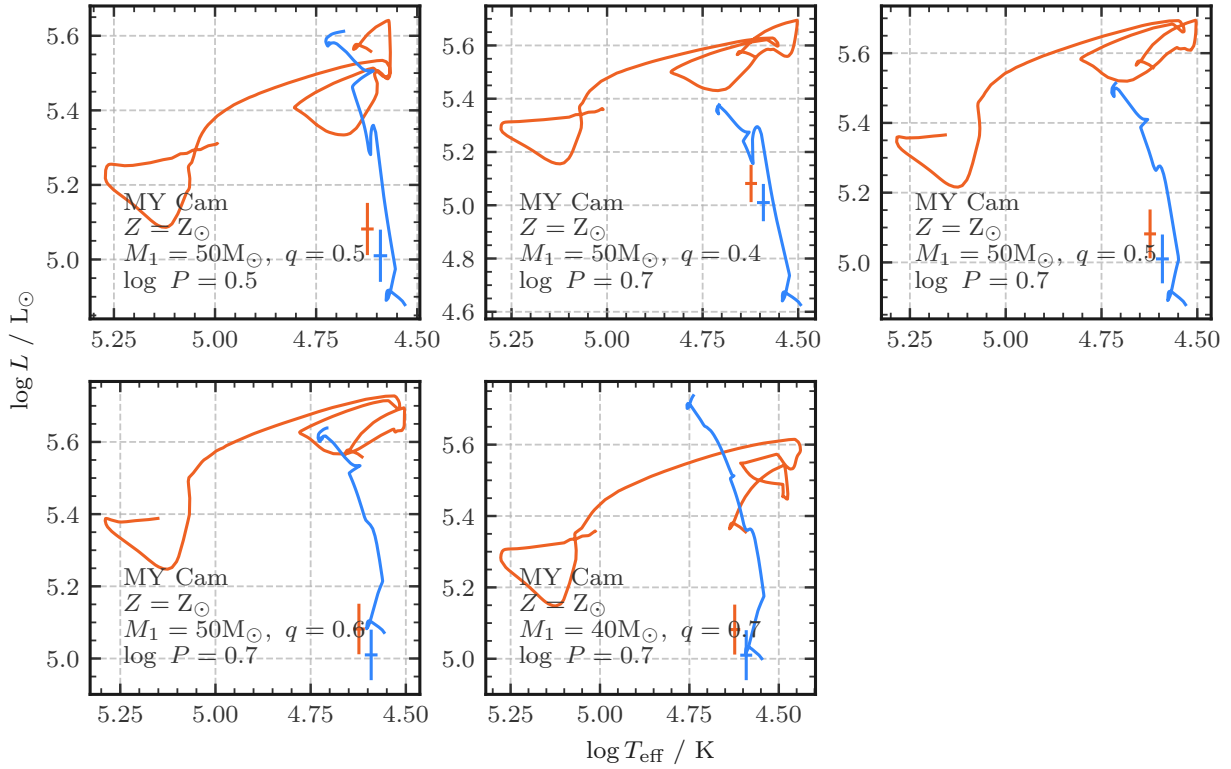


(f) VFTS 352

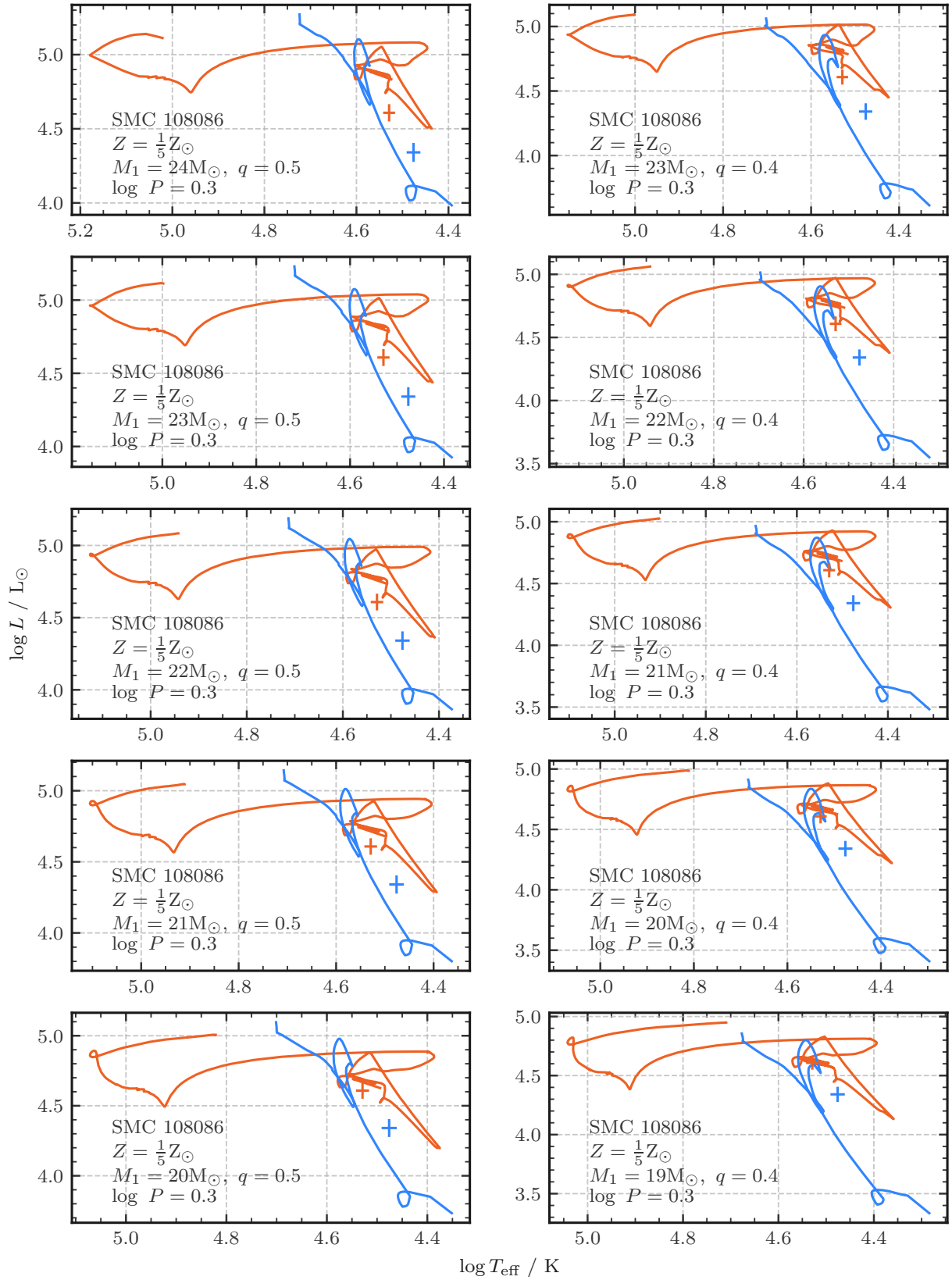
Figure D.2: The results of our bespoke binary evolution for **wider** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.



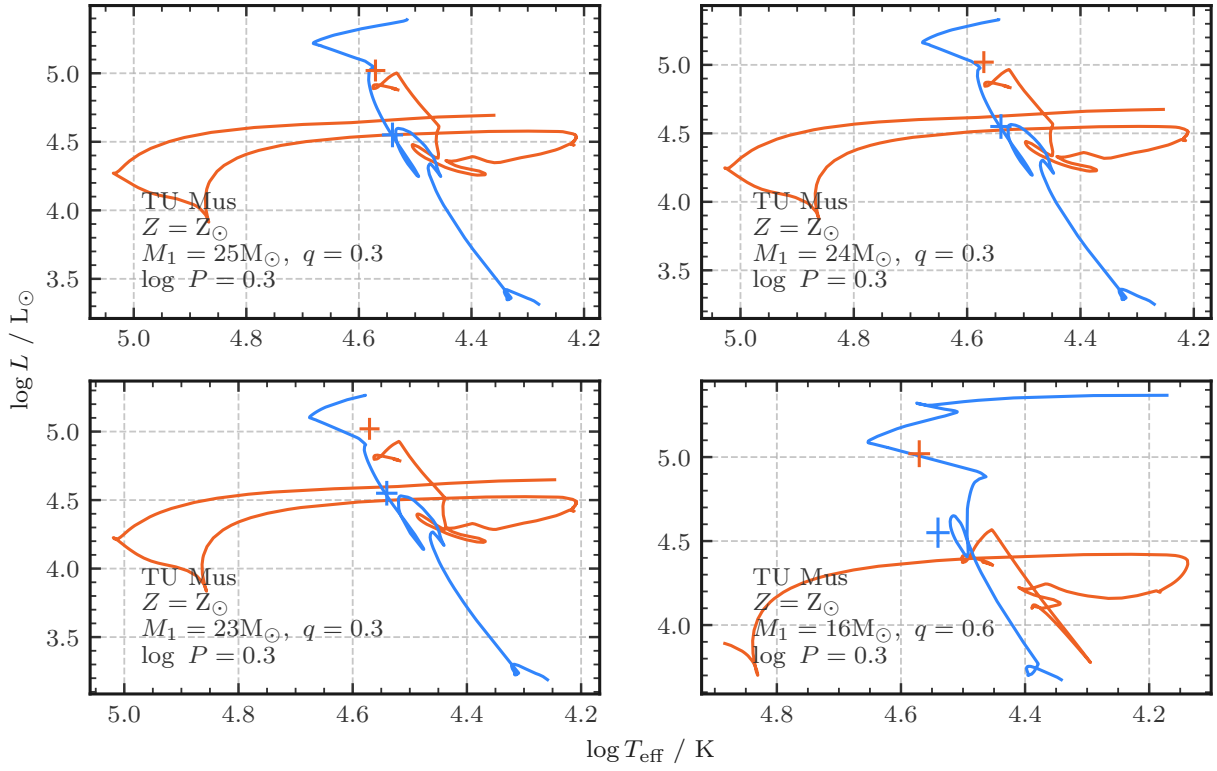
(a) LSS 3074



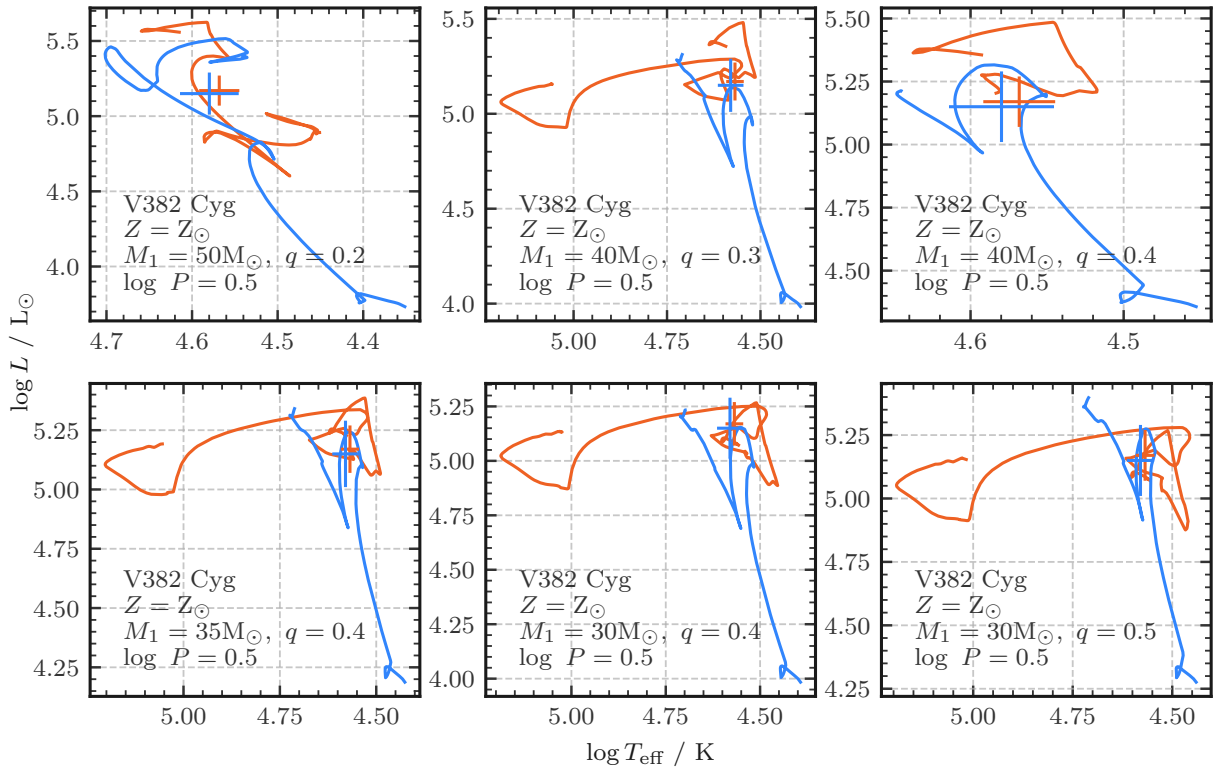
(b) MY Camelopardalis



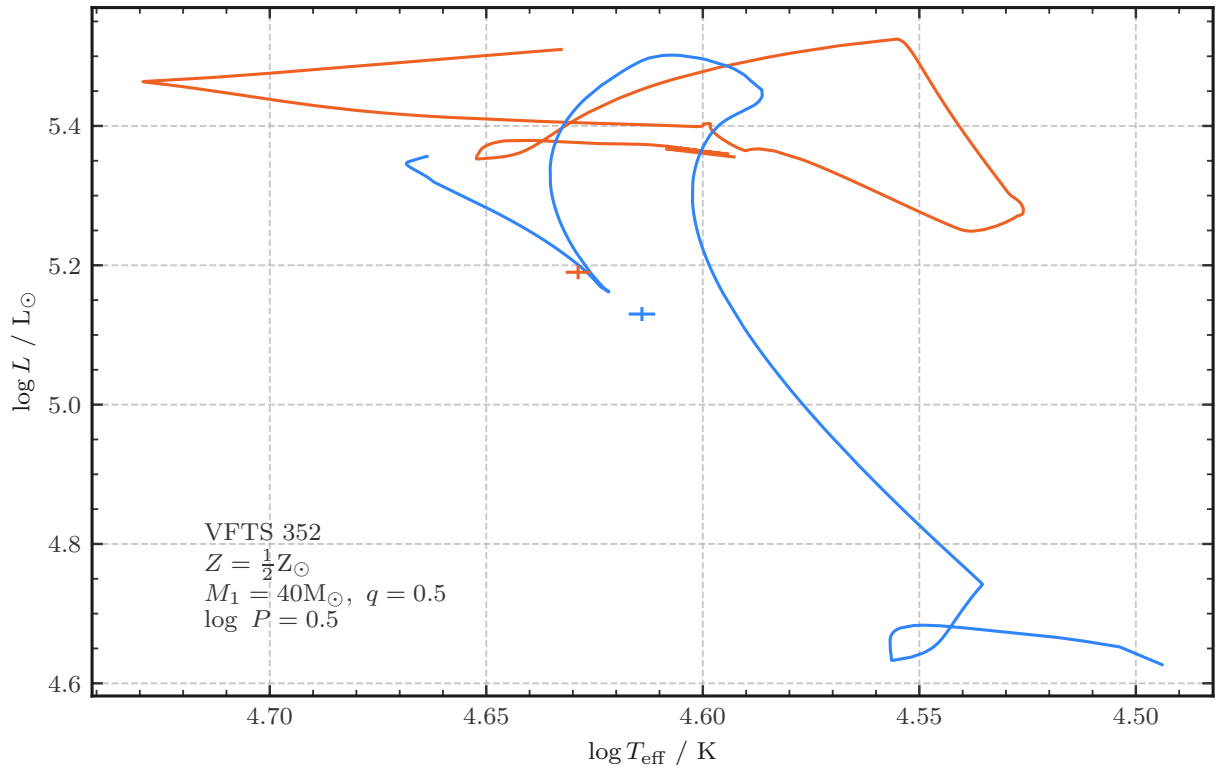
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure D.3: The results of our bespoke binary evolution for **shorter** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.

